

A modular–2-adic two-row approximation to Catalan’s constant with worthiness 0.9025266028 . . .

Abstract

Let $G = \beta(2)$ be Catalan’s constant and $\varphi = (1 + \sqrt{5})/2$. We combine Zudilin’s Apéry-like Catalan recurrence with a second, modular recurrence coming from a weight-3 Eisenstein series with character χ_{-4} . The key arithmetic input is that the two rational approximants converge to the same 2-adic Catalan period, but with different speeds. Sampling the Zudilin row at $5n$ and the modular row at $8n$ balances these two 2-adic speeds and produces a hidden determinant divisor of size $2^{40n+o(n)}$. A correlated rank-two congruence lattice then gives rational approximants p_n/q_n satisfying

$$\liminf_{n \rightarrow \infty} \frac{-\log |G - p_n/q_n|}{\log |q_n|} \geq \frac{50 \log \varphi}{10 - 2 \log 2 + \frac{75}{2} \log \varphi} = 0.902526602856971 \dots$$

The bound is below 1, so it is not an irrationality proof. We also explain why the ratio 5 : 8 is forced by the arithmetic and is optimal within this particular two-row scalar architecture.

1 The two Apéry rows

We write $D_N = \text{lcm}(1, \dots, N)$. Zudilin’s Catalan sequences Q_m, P_m are characterized by

$$Q_0 = 1, \quad Q_1 = \frac{7}{4}, \quad P_0 = 0, \quad P_1 = \frac{13}{8},$$

and the same second-order recurrence

$$\begin{aligned} & (2m+1)^2(2m+2)^2(20m^2 - 8m + 1)u_{m+1} \\ &= (3520m^6 + 5632m^5 + 2064m^4 - 384m^3 - 156m^2 + 16m + 7)u_m \\ &+ (2m-1)^2(2m)^2(20m^2 + 32m + 13)u_{m-1}. \end{aligned}$$

We use the standard asymptotics

$$\frac{1}{m} \log |Q_m| \rightarrow 5 \log \varphi, \quad \frac{1}{m} \log |Q_m G - P_m| \rightarrow -5 \log \varphi, \quad (1)$$

and Zudilin’s denominator bounds, in the convenient form

$$2^{e_m} Q_m \in \mathbf{Z}, \quad 2^{e_m} D_{2m-1}^2 P_m \in \mathbf{Z}, \quad e_m = 4m + O(\log m). \quad (2)$$

The second row is modular. Put

$$t(\tau) = \frac{\eta(\tau)^4 \eta(4\tau)^2 \eta(8\tau)^4}{\eta(2\tau)^{10}}, \quad F(\tau) = \frac{\eta(2\tau)^{10}}{\eta(\tau)^4 \eta(4\tau)^4}.$$

Then $F = \sum_{n \geq 0} A_n t^n$, where

$$(n+1)^2 u_{n+1} = (12n(n+1) + 4)u_n - 32n^2 u_{n-1}, \quad (3)$$

with $A_0 = 1, A_1 = 4$. The canonical modular source is

$$\Phi_E = F q \frac{dt}{dq} = S - 8S(2\tau),$$

where $S = E_{3, \chi_{-4}, 1}$ in Fourier normalization. Hence

$$L(\Phi_E, s) = (1 - 8 \cdot 2^{-s})\beta(s)\zeta(s-2), \quad L(\Phi_E, 2) = \frac{G}{2}.$$

This explains the choice of the modular form: weight 3 is the natural weight for the value $L(2, \chi_{-4})$, and the oldform factor $1 - 8V_2$ kills the unwanted endpoint at $s = 3$ while retaining the Catalan period.

Let $B(t) = \sum B_n t^n$ be the normalized companion $B_0 = 0, B_1 = 1$. The modular rectification is

$$B(t(q)) = F(q)D_q^{-2}\Phi_E(q), \quad D_q = q \frac{d}{dq}. \quad (4)$$

Two integrations immediately give

$$A_n \in \mathbf{Z}, \quad D_n^2 B_n \in \mathbf{Z}. \quad (5)$$

Indeed the coefficient of t^n only sees Fourier indices $m \leq n$, and D_q^{-2} contributes only m^{-2} . The characteristic roots of (3) are 8 and 4; therefore

$$\frac{1}{n} \log |A_n| \rightarrow \log 8, \quad \frac{1}{n} \log \left| \frac{G}{2} A_n - B_n \right| \rightarrow \log 4. \quad (6)$$

2 The hidden 2-adic bridge

The crucial fact is that the two recurrences represent the same 2-adic period. We isolate the exact statement needed below.

Lemma 1 (common 2-adic Catalan period). *There exists $G_2 \in \mathbf{Q}_2$ such that*

$$v_2 \left(G_2 - \frac{P_m}{Q_m} \right) = 8m - 1 - 4s_2(m), \quad (7)$$

$$v_2 \left(G_2 - \frac{2B_n}{A_n} \right) = 5n - 1 - 4s_2(n), \quad (8)$$

where $s_2(r)$ is the number of 1's in the binary expansion of r . Consequently

$$v_2(P_m A_{2m} - 2B_{2m} Q_m) = 4m - 1. \quad (9)$$

Idea of proof. The exact Wronskians are

$$P_m Q_{m-1} - Q_m P_{m-1} = (-1)^{m-1} \frac{20m^2 - 8m + 1}{8m^2(2m-1)^2}$$

and

$$A_n B_{n-1} - A_{n-1} B_n = -\frac{32^{n-1}}{n^2}.$$

Together with

$$v_2(Q_m) = -4m + 2s_2(m), \quad v_2(A_n) = 2s_2(n),$$

they give the exact tail valuations (7) and (8) once the two limits are identified.

That identification is the meeting point of two classical constructions. Rivoal's moving Padé specialization of Beukers's continued fraction is exactly Zudilin's sequence. If one defines the Padé target with the sign-safe normalization

$$\mathcal{T}(x) = 2R(x) - R(x/2),$$

where R is Beukers's p -adic Hurwitz series, then

$$\mathcal{T}(x+1) + \mathcal{T}(x) = \frac{2}{x^2}, \quad \mathcal{T}_2(1/2) = 8\zeta_2(2).$$

At the moving point $x_m = \frac{1}{2} - m$, Rivoal's exact Padé identity and the shift equation show that P_m/Q_m converges to the single constant $\zeta_2(2)$. Beukers's Padé estimate is uniform for the odd numerator $1 - 2m$, so the moving-point passage is legitimate.

On the modular side, (4) is the level-8 form of Calegari's overconvergent 2-adic Catalan construction: after the standard Teichmüller normalization its constant term is the same $\zeta_2(2)$. Thus both limits equal $G_2 := \zeta_2(2)$. Finally, at $n = 2m$ the two tail valuations are respectively

$$8m - 1 - 4s_2(m), \quad 10m - 1 - 4s_2(m),$$

so the first is strictly smaller. Ultrametricity therefore gives equality for the difference of the two ratios; multiplying by $Q_m A_{2m}$ yields (9). $\square$

The diagonal identity is more useful than it first appears, because it forces the two 2-adic limits to be equal; hence the exact tails give an off-diagonal inequality for arbitrary indices. If

$$\Delta_{i,k} = P_i A_k - 2Q_i B_k,$$

then

$$v_2(\Delta_{i,k}) \geq \min\{4i - 1 - 2s_2(i) + 2s_2(k), 5k - 4i - 1 + 2s_2(i) - 2s_2(k)\}. \quad (10)$$

The two affine slopes balance when $8i = 5k$. This is the arithmetic reason for the apparently strange sampling ratio

$$\boxed{i = 5n, \quad k = 8n.}$$

Since $s_2(8n) = s_2(n)$, (10) gives

$$v_2(\Delta_{5n,8n}) \geq 20n - 1 - 2|s_2(5n) - s_2(n)| = 20n - O(\log n). \quad (11)$$

3 The correlated congruence lattice

Put

$$S_n = D_{10n}^2.$$

Since $10n$ is never a prime power, $D_{10n} = D_{10n-1}$. Define the two integer rows

$$(X_{1,n}, Y_{1,n}) = S_n(A_{8n}, 2B_{8n}),$$

$$(X_{2,n}, Y_{2,n}) = 2^{e_{5n}} S_n(Q_{5n}, P_{5n}).$$

Write $X_{r,n} = S_n a_{r,n}$. Their mixed minor after removing the common S_n from the first coordinates is

$$h_n = a_{1,n} Y_{2,n} - a_{2,n} Y_{1,n} = 2^{e_{5n}} S_n \Delta_{5n,8n}.$$

Since $e_{5n} = 20n + O(\log n)$ and (11) holds, there is an explicit power T_n with

$$T_n \mid h_n, \quad \frac{1}{n} \log T_n \rightarrow 40 \log 2. \quad (12)$$

For example one may take

$$T_n = 2^{40n-3-2\lfloor \log_2(5n) \rfloor}.$$

The point is that this determinant divisibility is *correlated*: it can be used in two congruences at the cost of one copy of T_n , rather than T_n^2 . Set $M_n = S_n T_n$ and consider

$$K_n = \left\{ (c_1, c_2) \in \mathbf{Z}^2 : \begin{array}{l} a_{1,n}c_1 + a_{2,n}c_2 \equiv 0 \pmod{T_n}, \\ Y_{1,n}c_1 + Y_{2,n}c_2 \equiv 0 \pmod{S_n T_n} \end{array} \right\}.$$

Then

$$q(c) = \frac{c_1 X_{1,n} + c_2 X_{2,n}}{M_n}, \quad p(c) = \frac{c_1 Y_{1,n} + c_2 Y_{2,n}}{M_n}$$

are integers.

A two-line Smith-normal-form computation gives the useful index formula

$$[\mathbf{Z}^2 : K_n] = \frac{S_n T_n^2}{\gcd(h_n, T_n Y_{1,n}, T_n Y_{2,n}, S_n T_n a_{1,n}, S_n T_n a_{2,n}, S_n T_n^2)}.$$

Since $T_n \mid h_n$, we obtain

$$[\mathbf{Z}^2 : K_n] \leq S_n T_n = M_n. \quad (13)$$

By replacing one basis vector by a suitable positive integer multiple we may, without losing integrality, pass to a sublattice $L_n \subseteq K_n$ with

$$M_n \leq \text{covol}(L_n) < 2M_n. \quad (14)$$

Thus the division modulus and the coefficient-lattice covolume have the same exponential rate

$$\sigma = 20 + 40 \log 2, \quad (15)$$

because $\log D_N = N + o(N)$.

4 Balancing the two rows

Let

$$\lambda_{r,n} = X_{r,n}G - Y_{r,n}.$$

From (1)–(6) and $\log S_n = 20n + o(n)$ we get

$$\begin{aligned} A_1 &= 20 + 24 \log 2, & E_1 &= 20 + 16 \log 2, \\ A_2 &= 20 + 20 \log 2 + 25 \log \varphi, & E_2 &= 20 + 20 \log 2 - 25 \log \varphi, \end{aligned}$$

where A_r is the exponential height of $X_{r,n}$ and E_r that of $\lambda_{r,n}$. Moreover

$$(A_2 + E_1) - (A_1 + E_2) = 50 \log \varphi - 8 \log 2 > 0,$$

so the mixed determinant has a unique dominant real term.

We use the following standard rank-two form of the two-successive-minima argument. If a coefficient lattice has covolume $e^{\sigma n + o(n)}$, the outputs are divided by $e^{\sigma n + o(n)}$, and the above crossed gap holds, then choose the coefficient anisotropy

$$x = \frac{\sigma + E_2 - E_1}{2}.$$

Minkowski's second theorem applied to the rectangle $|c_1| \leq e^{xn}$, $|c_2| \leq e^{(\sigma-x)n}$ yields an integer output with

$$\log |q_n| \geq Hn - o(n), \quad \log |q_n G - p_n| \leq Fn + o(n), \quad (16)$$

where

$$H = A_2 - x, \quad F = x + E_1 - \sigma.$$

The use of two successive minima, rather than only Minkowski's first theorem, is what prevents an exceptionally short rationality-kernel vector from spoiling the lower bound on $|q_n|$.

Substituting (15) gives

$$x = 10 + 22 \log 2 - \frac{25}{2} \log \varphi,$$

and therefore

$$\boxed{H = 10 - 2 \log 2 + \frac{75}{2} \log \varphi}, \quad \boxed{F = 10 - 2 \log 2 - \frac{25}{2} \log \varphi}. \quad (17)$$

Both are positive, and $H - F = 50 \log \varphi$.

Theorem 2. *There are integer pairs (q_n, p_n) with $q_n \neq 0$, $q_n G - p_n \neq 0$ and $|q_n| \rightarrow \infty$ such that*

$$\liminf_{n \rightarrow \infty} \frac{-\log |G - p_n/q_n|}{\log |q_n|} \geq \frac{50 \log \varphi}{10 - 2 \log 2 + \frac{75}{2} \log \varphi} = 0.9025266028569714694 \dots$$

Proof. Apply (16) and use

$$-\log |G - p_n/q_n| = \log |q_n| - \log |q_n G - p_n|.$$

Hence the liminf is at least $1 - F/H = (H - F)/H$, and (17) gives the stated constant. $\square$

5 Why 5 : 8 is the scalar optimum

There is a useful conceptual coda. Sample Zudilin at $i = an$ and the modular row at $k = cn$. The common LCM-square contributes $2 \max(2a, c)$ to the exponential logarithmic cost, while the 2-adic roof (10) contributes

$$\min(8a, 5c) \log 2.$$

The same balancing calculation gives

$$H(a, c) = \max(2a, c) + (2a + c) \log 2 - \frac{1}{2} \min(8a, 5c) \log 2 + \frac{15}{2} a \log \varphi,$$

and

$$\delta(a, c) = \frac{10a \log \varphi}{H(a, c)}.$$

By homogeneity put $r = c/a$. Then $H(r)$ is strictly decreasing for $0 < r < 8/5$ and strictly increasing for $r > 8/5$ (with the elementary change of formula at $r = 2$). Thus

$$\boxed{r = \frac{8}{5}}$$

is the unique optimum within this specific scalar architecture. In other words, the same equality $8i = 5k$ that balances the two 2-adic convergence rates also optimizes the final Diophantine exponent.

Remark 3. *The theorem is deliberately a statement about the worthiness of an explicit family of rational approximations. Since the bound is < 1 , it does not imply that G is irrational. The scalar optimum above is likewise only an optimum inside the present two-realization, common-LCM-square, 2-adic-cross, two-minimum construction; it is not a universal obstruction to other Catalan constructions.*

References

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